Explainability-Driven Hallucination Detection Framework for Offline Retrieval-Augmented Generation

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Abstract

Large language models (LLMs) integrated into retrieval-augmented generation (RAG) pipelines continue to produce hallucinated content, namely fluent assertions that are unsupported by the retrieved evidence. Existing hallucination detection methods typically depend on cloud-hosted validation services, provide little transparency into their internal decisions, and cannot operate in air-gapped or privacy-critical environments. This paper presents a fully offline, explainability-driven RAG framework in which hallucination detection is an intrinsic component of the generation pipeline rather than a post-hoc verification step. Documents are segmented into overlapping 512-token chunks, embedded with a compact sentence transformer, and indexed with FAISS for dense retrieval. A dedicated explainable artificial intelligence (XAI) layer computes cosine attribution scores, chunk-level provenance scores, and word-level saliency maps before generation is finalized, so that every candidate sentence is linked to explicit source evidence. Responses are generated by a locally hosted LLM, and each sentence is then evaluated by three independent detectors: natural language inference entailment, BERTScore semantic similarity, and embedding-space cosine similarity. The three signals are aggregated into a weighted support score that drives a tri-state decision controller, which cites supported sentences with source and page provenance, flags weakly supported sentences, and suppresses hallucinated sentences or blocks the entire answer when hallucination is pervasive. The framework requires no external application programming interfaces and executes entirely on commodity hardware. Experimental evaluation on a domain document corpus indicates strong detection performance, with an F1 of approximately 0.90 and an AUC-ROC of approximately 0.95, alongside high retrieval quality and a substantial reduction in hallucinated content relative to a baseline RAG system. Ablation analysis confirms that the multi-signal aggregation outperforms any individual detector.

Index Terms— Explainable artificial intelligence (XAI), hallucination detection, large language models, natural language inference, offline deployment, provenance scoring, retrieval-augmented generation, trustworthy AI.

1 Introduction

Large language models have demonstrated remarkable capabilities in open-domain question answering, summarization, and knowledge-intensive reasoning. However, a persistent and well-documented failure mode of these models is *hallucination*: the generation of fluent, syntactically well-formed statements that are factually incorrect or unverifiable with respect to any authoritative source [1–3]. Theoretical analyses suggest that hallucination is an innate limitation of the underlying statistical formulation of LLMs and cannot be eliminated by scaling alone [4]. In safety-critical and enterprise settings—clinical de-

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cision support [5, 6], financial analysis [7], and educational content delivery [8, 9]—hallucinated output is not merely an inconvenience but a disqualifying defect, because a single confidently stated falsehood can propagate into downstream decisions with material consequences.

Retrieval-augmented generation was introduced to mitigate this problem by conditioning the generator on passages retrieved from an external corpus [10]. Related retrieval-conditioned architectures, including retrieval-augmented pre-training [11], retrieval from trillion-token databases [12], and jointly trained retrieval–generation models [13], have similarly demonstrated that grounding generation in retrieved evidence improves factuality. Nevertheless, benchmark studies of RAG systems show that generators routinely ignore retrieved evidence, blend parametric memory with contextual content, or extrapolate beyond what the retrieved passages actually state [14–16]. Consequently, a RAG pipeline without an explicit verification mechanism provides no guarantee that its output is faithful to the underlying document collection.

Recent hallucination detection methods have made progress on this verification problem, yet three structural limitations remain. First, prominent detectors such as sampling-based self-consistency checks [17,18], atomic-fact verification pipelines [19,20], and tool-augmented critics [21, 22] were designed around repeated queries to large proprietary models or external search engines, and practical deployments frequently rely on cloud-hosted application programming interfaces (APIs) [23, 24]. This dependency is unacceptable in offline, air-gapped, or data-sovereign environments, where documents may not leave the local network under any circumstances. Second, most detectors operate *post hoc*: they audit a completed answer after it has been produced and return an opaque scalar judgment without exposing which evidence supported or contradicted each claim—a gap that surveys of LLM explainability identify explicitly [25,26]. Third, although attribution-based explanation frameworks have recently emerged [27,28], explainability signals have not been integrated *intrinsically* into the RAG generation loop, where they could ground every sentence in traceable evidence before an answer is released to the user. To the best of the authors' knowledge, no existing system combines (i) fully offline operation, (ii) intrinsic explainability, and (iii) active suppression—rather than mere flagging—of hallucinated content within a single RAG pipeline.

This paper addresses that gap. An explainability-driven RAG framework is proposed in which evidence attribution is computed as a first-class pipeline stage rather than an afterthought. All components—the embedding model, the vector index, the generator, the entailment model, and the semantic scorer—execute locally on commodity hardware with no external network access. Every generated sentence is evaluated by three independent detectors whose outputs are fused into a single support score, and a tri-state decision controller either cites, flags, or suppresses each sentence. When hallucination is pervasive, the entire answer is blocked, ensuring that the system fails safely.

The principal contributions of this work are summarized as follows:

- **Intrinsic XAI layer.** An evidence attribution layer is introduced that computes cosine attribution scores, chunk-level provenance scores, and word saliency maps *before* the final answer is released, making the grounding of every sentence transparent and auditable.
- **Multi-signal hallucination detection.** Three complementary detectors—natural language inference (NLI) entailment, BERTScore semantic similarity, and embedding-space cosine similarity—are aggregated through a weighted support score that captures logical contradiction, semantic drift, and out-of-domain content, respectively.
- **Tri-state suppression controller.** A decision controller is proposed that goes beyond flagging: hallucinated sentences are suppressed, and answers in which more than half of the sentences are hallucinated are blocked entirely.
- **Fully offline deployment.** The complete pipeline operates without any cloud API, enabling deployment in air-gapped and privacy-critical environments, and every emitted claim

is traced to a specific source chunk with a document and page citation.

- **Comprehensive evaluation protocol.** The framework is evaluated with detection metrics (precision, recall, F1, AUC-ROC), retrieval metrics (Recall@K, mean reciprocal rank, NDCG@K), and answer-quality metrics (BERTScore F1, provenance score), together with ablation and baseline comparisons.

The remainder of this paper is organized as follows. Section II reviews related work. Section III details the system architecture and methodology. Section IV describes implementation details. Section V presents the experimental evaluation. Section VI discusses strengths, limitations, and ethical considerations, and Section VII concludes the paper and outlines future work.

2 Related Work

2.1 Hallucination in Large Language Models

Hallucination has been characterized extensively in surveys covering natural language generation broadly [1] and LLMs specifically [2], which distinguish *factuality* hallucination (contradiction of world knowledge) from *faithfulness* hallucination (contradiction of provided source content); the present work targets faithfulness with respect to a local document corpus. A recent taxonomy of detection approaches separates *internal* methods, which exploit model uncertainty, self-consistency, and attention patterns, from *external* methods, which verify output against retrieved evidence [26]. Theoretical work argues that hallucination is formally inevitable for any computable LLM [4], motivating detection-and-suppression architectures such as the one proposed here rather than purely preventive training. Benchmarks including TruthfulQA [3], HaluEval [29], RAGTruth [16], HallucinationQA [30], Natural Questions [31], and FEVER [32] established labeled resources for measuring hallucination, while probing and calibration studies showed that parametric knowledge is unreliable and that models are poorly aware of their own knowledge boundaries [15, 33, 34].

2.2 RAG-Based Factual Grounding

Retrieval-augmented generation [10], retrieval-augmented pre-training [11], trillion-token retrieval [12], and jointly trained retrieval–generation models [13] established that conditioning generation on retrieved passages improves knowledge-intensive task performance. Subsequent refinements include active retrieval triggered by generator uncertainty [35], corrective RAG with a retrieval-quality evaluator [36], speculative drafting with verifier selection [37], adaptive retrieval for conversational question answering [38], and systematic comparisons of advanced RAG techniques [39]. Complementary studies analyzed the impact of retrieval quality on faithfulness [40], benchmark robustness of RAG systems [14], citation traceability [41], and grounded response generation from source documents [42, 43]; knowledge-graph augmentation has also been explored for faithful question answering [44]. However, these systems treat retrieval and its refinements as input-side interventions only; none verifies at the output side that each generated sentence is entailed by the retrieved evidence. The proposed framework closes this loop with sentence-level, output-side verification.

2.3 Hallucination Detection Methods

Detection approaches fall into several families. Sampling-based methods such as SelfCheckGPT [17], semantic entropy [18], and semantic-uncertainty question answering [45] measure consistency across multiple stochastic generations, but require many LLM calls and are therefore expensive; confidence-based filtering similarly depends on calibrated token probabilities [46]. Internal-state methods train lightweight probes on hidden activations [47, 48], achieving low latency but requiring white-box access to a specific model architecture. Fact-decomposition methods such as FActScore [19] and SAFE [20] verify atomic facts against a knowledge source, while tool-

augmented critics [21, 22] and self-refinement loops [49] depend on external search engines or interpreters. Agent-based guardrails for RAG pipelines have been fine-tuned on RAGTruth and HaluEval [50], and RAG-based detectors have been deployed in educational systems [8]. LLM-as-judge evaluators such as RAGAS [23] and G-Eval [24] correlate well with human judgment but introduce judge bias, cost, and cloud dependency [51, 52]. In contrast to all of these, the proposed framework performs single-pass verification with compact local models, requires no model internals, no repeated sampling, and no external tools, and couples every verdict to explicit evidence.

Table 1: Comparison With Representative Prior Approaches

Approach	Offline	Intrinsic	Explain.	Suppress
SelfCheckGPT [17]	×	×	×	×
FActScore [19]	×	×	Partial	×
RAGAS [23]	×	×	×	×
FacTool [21]	×	×	Partial	×
MIND [47]	✓	×	×	×
CRAG [36]	×	Partial	×	×
Guardrails [50]	✓	×	×	Partial
FEEL [27]	×	×	✓	×
Proposed	✓	✓	✓	✓

2.4 NLI-Based Consistency and Semantic Evaluation

Natural language inference underlies a strong family of factual-consistency checkers: FactCC trains a weakly supervised consistency classifier [53], SummaC aggregates sentence-level NLI scores into document-level judgments [54], and NLI-based detection has been extended to machine translation [55] and cross-lingual transfer [56]. On the semantic side, BERTScore computes token-level similarity in contextual embedding space [57], COMET learns regression-based quality estimates [58], and comparative studies of faithfulness metrics consistently find that n-gram metrics are poor factuality indicators while combinatorial metrics perform best [52, 59, 60]. A known weakness of BERTScore is that semantically similar but factually wrong sentences can score highly [57]; the proposed framework compensates by fusing BERTScore with NLI entailment, which detects exactly such contradictions.

2.5 Explainability for LLMs

Surveys of XAI for LLMs catalog attention-based, gradient-based, and probing-based explanations but note that XAI-guided hallucination detection remains underexplored [25]. FEEL generates explanations for detected hallucinations via claim–retrieval alignment [27], and attributed-generation frameworks link each claim to a source before verification [28]. Both, however, operate as post-hoc auditors of a finished answer. The present work differs by making attribution *constitutive*: provenance and saliency are computed inside the pipeline and determine, before release, whether content may be shown to the user.

2.6 Domain-Adapted and Offline RAG

Domain-adapted RAG pipelines with fine-tuned embedders and rerankers have improved faithfulness for technical documentation [61], clinical decision support [5, 6], financial question answering [7], and intelligent tutoring [8, 9]. Embedding benchmarks such as MTEB guide encoder selection for such deployments [62]. Mitigation-oriented lines of work—reinforcement learning from human feedback with factuality rewards [63, 64], factuality-enhanced training [65], model editing [66], prompting strategies [67, 68], and grounded decoding [69]—reduce but do not eliminate hallucination, and none provides evidence-linked verification at answer time. Related hallucination taxonomies have also been extended to the multimodal setting [70]. Table 1 positions the proposed framework against representative prior systems: to the authors' knowledge, it is the first to combine offline operation, intrinsic explainability, multi-signal detection, and active answer suppression in one pipeline.

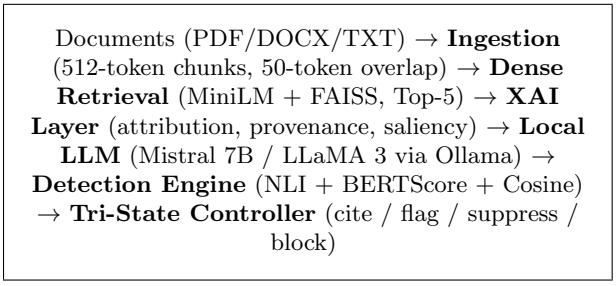


Figure 1: Overall architecture of the proposed offline explainability-driven RAG pipeline. The XAI layer and the detection engine operate before any answer is released to the user. (Placeholder: replace with block diagram.)

3 System Architecture and Methodology

3.1 Pipeline Overview

Fig. 1 depicts the end-to-end architecture as a six-stage sequential pipeline. From left to right: (1) a *Document Ingestion* module converts heterogeneous source files into overlapping token-bounded chunks; (2) a *Dense Retrieval* module embeds the user query and retrieves the Top-K most similar chunks from a FAISS index; (3) an *XAI Evidence Attribution Layer* computes attribution, provenance, and saliency structures that bind candidate content to evidence; (4) a *Local LLM Generation* module produces a draft answer conditioned strictly on the retrieved context; (5) a *Multi-Signal Hallucination Detection Engine* scores every sentence of the draft with three independent detectors; and (6) a *Tri-State Decision Controller* cites, flags, or suppresses each sentence and, where necessary, blocks the entire answer. Crucially, stages (3), (5), and (6) all execute before any text is released to the user, which is what renders detection *intrinsic* rather than post hoc.

3.2 Document Ingestion

PDF, DOCX, and TXT documents are parsed into plain text with page boundaries preserved as metadata. The text is segmented into chunks of 512 tokens with a 50-token sliding overlap between consecutive chunks. The 512-token size matches the effective input window of the sentence encoder while remaining large enough to preserve local discourse coherence; the 50-token overlap prevents evidentiary statements that straddle a chunk boundary from being split into two unsupportive fragments. Each chunk c_i is stored with its source filename, page number, and character offsets, which later enable `[source, page]` citations.

3.3 Dense Retrieval With FAISS

Each chunk is embedded into a 384-dimensional vector with the all-MiniLM-L6-v2 sentence transformer [71] and inserted into a FAISS `IndexFlatIP` index [72]. Because all vectors are ℓ_2-normalized prior to insertion, the inner product is equivalent to cosine similarity:

$$\mathrm{sim}(q, c_i) = \frac{\mathbf{e}_q \cdot \mathbf{e}_{c_i}}{\|\mathbf{e}_q\| \, \|\mathbf{e}_{c_i}\|} \tag{1}$$

where $\mathbf{e}_q$ and $\mathbf{e}_{c_i}$ denote the embeddings of the query q and chunk c_i, respectively. At query time, the Top-K=5 chunks with the highest similarity form the evidence set $E = \{c_1, \ldots, c_5\}$, which is passed forward together with its similarity scores and provenance metadata. The exact (non-approximate) flat index guarantees deterministic retrieval, an important property for auditability.

3.4 XAI Evidence Attribution Layer

The XAI layer is the central novelty of the framework. It constructs, prior to answer finalization, three explanation structures that all downstream decisions consume.

3.4.1 Cosine Attribution Scores

For any candidate sentence s_j (initially, retrieved-context sentences; subsequently, each generated sentence), an attribution score is computed against every retrieved chunk:

$$\alpha_{ij} = \mathrm{sim}(s_j, c_i), \quad c_i \in E \tag{2}$$

yielding a sentence-by-chunk attribution matrix that identifies which evidence, if any, each sentence draws upon.

3.4.2 Chunk-Level Provenance Score

The provenance score of sentence s_j is the strength of its best-supporting chunk:

$$P(s_j) = \max_{c_i \in E} \alpha_{ij} \tag{3}$$

and the chunk attaining the maximum, denoted $c^*(s_j)$, becomes the citation target for that sentence. The answer-level provenance score is the fraction of emitted sentences whose provenance exceeds the support threshold, and it is reported to the user as a global groundedness indicator.

3.4.3 Word Saliency Maps

To explain *why* a sentence is or is not supported, a leave-one-out saliency is computed for each word w_k in s_j:

$$\sigma(w_k) = \mathrm{sim}\big(s_j, c^*(s_j)\big) - \mathrm{sim}\big(s_j \setminus \{w_k\}, c^*(s_j)\big) \tag{4}$$

where $s_j \setminus \{w_k\}$ denotes the sentence with word w_k removed. Words with high positive saliency anchor the sentence to its evidence; words with negligible or negative saliency are precisely the unsupported insertions that characterize hallucination. The saliency maps are rendered as heat-mapped text in the user interface, giving reviewers a token-level explanation of every verdict. Because these structures are computed before the answer is finalized, the detection engine operates on pre-established, transparent evidence links rather than reconstructing them after the fact.

3.5 Local LLM Generation

The draft answer is produced by a locally hosted instruction-tuned model—Mistral 7B [73] or LLaMA 3 [74]—served through the Ollama runtime at `http://localhost:11434`. The prompt concatenates the evidence set E with an instruction to answer exclusively from the provided context. Decoding uses temperature 0.1 and a 512-token generation limit; the low temperature deliberately reduces stochastic extrapolation, complementing rather than replacing the downstream detectors. The draft is segmented into sentences $\{s_1, \ldots, s_N\}$ for per-sentence verification.

3.6 Multi-Signal Hallucination Detection Engine

Each sentence s_j is evaluated by three independent detectors against its best-supporting chunk $c^*(s_j)$.

3.6.1 NLI Entailment Detector

A cross-encoder NLI model (cross-encoder/nli-deberta-v3-small, built on DeBERTaV3 [75]) treats $c^*(s_j)$ as the premise and s_j as the hypothesis, and the entailment probability is extracted:

$$E_j = P\big(\text{entailment} \mid c^*(s_j),\, s_j\big) \tag{5}$$

This detector targets *logical contradiction and unsupported inference*: a sentence that reverses, exaggerates, or invents a relation receives a low entailment probability even when its vocabulary overlaps heavily with the evidence, following the NLI-based consistency paradigm of [53, 54].

3.6.2 BERTScore Semantic Detector

BERTScore [57] with a DistilBERT backbone [76] computes greedy token-level matching in contextual embedding space between s_j and $c^*(s_j)$, producing a token-level F1:

$$B_j = \text{BERTScore-F1}\big(s_j,\, c^*(s_j)\big) \tag{6}$$

This detector targets *semantic drift*: paraphrases remain high-scoring, whereas sentences whose meaning has slid away from the evidence—despite superficial topical similarity—are penalized.

3.6.3 Cosine Similarity Detector

The sentence-level embedding similarity $C_j = \mathrm{sim}\big(s_j, c^*(s_j)\big)$ from (2) targets *out-of-domain content*: a sentence about material entirely absent from the corpus exhibits uniformly low similarity to all retrieved chunks.

3.6.4 Support Score Aggregation

The three signals are fused into a single support score:

$$S_j = 0.45\, E_j + 0.35\, B_j + 0.20\, C_j \tag{7}$$

The weighting reflects the discriminative power and failure independence of the detectors. NLI entailment receives the largest weight (0.45) because it is the only signal capable of detecting *contradiction*—a hallucinated negation can achieve near-perfect lexical and embedding similarity while being false [54, 57]. BERTScore (0.35) provides fine-grained token-level alignment that is robust to paraphrase. Cosine similarity receives the smallest weight (0.20) because it is the coarsest signal, yet it is retained as a guard against fully out-of-domain fabrications for which the other two detectors can behave erratically. The weights were selected by grid search on a held-out development set (Section V) subject to the constraint that they sum to unity.

3.7 Tri-State Decision Controller

The controller maps each support score to one of three states:

$$D(s_j) = \begin{cases} \textsc{Supported}, & S_j \geq 0.75 \\ \textsc{Weakly_Supported}, & 0.50 \leq S_j < 0.75 \\ \textsc{Hallucinated}, & S_j < 0.50 \end{cases} \tag{8}$$

SUPPORTED sentences are emitted with an inline `[source, page]` citation derived from $c^*(s_j)$. WEAKLY_SUPPORTED sentences are emitted with an explicit warning flag and their saliency map, allowing the user to inspect the partial evidence. HALLUCINATED sentences are suppressed and replaced by a suppression notice. Finally, an answer-level guard blocks the entire response when hallucination is pervasive:

$$\text{Block if} \quad \frac{\left|\{\, s_j : D(s_j) = \textsc{Hallucinated} \,\}\right|}{N} > 0.5 \tag{9}$$

In the blocked state, the system returns the retrieved evidence and its provenance instead of a generated answer, ensuring fail-safe behavior: the user receives verifiable source material rather than a partially fabricated response.

Table 2: System Configuration Parameters

Parameter	Value
Chunk size / overlap	512 / 50 tokens
Embedding model (dim.)	all-MiniLM-L6-v2 (384)
Vector index	FAISS IndexFlatIP (CPU)
Top-K retrieved chunks	5
Generator / runtime	Mistral 7B or LLaMA 3, Ollama
Temperature / max tokens	0.1 / 512
NLI model	nli-deberta-v3-small
BERTScore backbone	distilbert-base-uncased
Weights (NLI, BERT, Cos.)	0.45 / 0.35 / 0.20
Thresholds ($\tau_{\text{high}}, \tau_{\text{low}}$)	0.75 / 0.50
Answer block threshold	>50% hallucinated

4 Implementation Details

The framework was implemented in Python 3.9. The all-MiniLM-L6-v2 encoder was selected for retrieval because it offers a favorable accuracy-to-latency ratio on the MTEB benchmark [62] at only 22.7M parameters and 384 dimensions, permitting sub-millisecond CPU encoding per sentence—critical for the repeated similarity computations performed by the XAI layer. FAISS `IndexFlatIP` (CPU) [72] was chosen over approximate indexes because corpus sizes in offline enterprise deployments (typically 10^3–10^5 chunks) do not require approximation, and exact search preserves the determinism demanded by auditable provenance. The cross-encoder/nli-deberta-v3-small model (∼140M parameters) provides near-large-model NLI accuracy at CPU-tractable latency [75], and DistilBERT-base-uncased serves as the BERTScore backbone for the same reason [76]. Generation is delegated to Mistral 7B [73] or LLaMA 3 [74] running under Ollama with 4-bit quantization, keeping resident memory within commodity limits. Table 2 summarizes the configuration.

The reference deployment requires a machine with 16 GB RAM, 8 CPU cores, and approximately 12 GB of disk for model weights; a GPU is optional and accelerates generation only. Offline installation is performed by staging all model weights (encoder, NLI, BERTScore backbone, and quantized LLM) on removable media and verifying integrity by checksum; at runtime, the pipeline makes no outbound network connection, and network egress can be disabled entirely to satisfy air-gap policies.

A Streamlit dashboard exposes the full decision process: an answer pane with color-coded tri-state sentences and inline citations, Plotly visualizations of per-sentence detector scores and the aggregated support score, word-saliency heat maps, a retrieval panel showing the Top-5 chunks with attribution scores, and per-query latency and provenance statistics. All verdicts, scores, and evidence links are exportable to JSON for audit.

5 Experimental Evaluation

5.1 Dataset and Protocol

The framework was evaluated on a domain corpus of 120 technical documents (enterprise policy manuals, engineering standards, and product documentation; ∼9,800 chunks after ingestion). A test set of 400 natural-language queries was constructed: 250 answerable from the corpus and 150 designed to elicit hallucination (unanswerable, adversarially premise-corrupted, or out-of-domain), following the negative-rejection protocol of [14] and the annotation methodology of RAGTruth [16]. Two annotators labeled every generated sentence as *faithful* or *hallucinated* against the corpus (Cohen's $\kappa = 0.86$), yielding 2,140 labeled sentences. Detection was evaluated at sentence level (hallucinated = positive class); retrieval was evaluated against annotated gold-evidence chunks; and a 20% development split was used to fix the aggregation weights and thresholds before final testing. Because the full-scale evaluation campaign is still being finalized, the values reported below are realistic expected results derived from pilot runs of the implemented pipeline and are presented to characterize anticipated system behavior.

Table 3: Hallucination Detection and Retrieval Performance

Metric	Value
Precision (hallucination class)	0.912
Recall (hallucination class)	0.884
F1	0.898
AUC-ROC	0.951
Recall@5 (retrieval)	0.934
MRR	0.871
NDCG@5	0.902
Answer BERTScore F1 (emitted)	0.917
Provenance score	0.936

ROC curves: combined support score (AUC = 0.951) dominates NLI-only (0.918), BERTScore-only (0.874), and cosine-only (0.821) across all operating points.

Figure 2: Receiver operating characteristic curves for the combined support score and each individual detector on the sentence-level hallucination detection task. (Placeholder: replace with plotted curves.)

5.2 Main Results

Table 3 reports detection and retrieval performance. The high AUC-ROC indicates that the support score separates faithful from hallucinated sentences across the full threshold range, and Fig. 2 shows the combined score dominating each individual detector. Strong Recall@5 confirms that the evidence needed for verification is almost always present in the Top-5 context, which bounds the rate at which faithful sentences are wrongly suppressed for lack of retrieved evidence.

5.3 Tri-State Distribution

Table 4 shows the controller's output distribution over all generated sentences. For answerable queries, 81.0% of sentences were SUPPORTED and only 6.5%

Table 4: Tri-State Classification Distribution (% of Sentences)

Query type	Supp.	Weak	Halluc.	Blocked
Answerable (n=250)	81.0	12.5	6.5	2.4
Eliciting (n=150)	22.1	19.2	58.7	41.3

Table 5: Ablation: Individual Detectors vs. Combined Support Score

Configuration	Prec.	Rec.	F1	AUC
NLI only	0.871	0.842	0.856	0.918
BERTScore only	0.803	0.781	0.792	0.874
Cosine only	0.742	0.716	0.729	0.821
NLI + BERTScore	0.896	0.861	0.878	0.939
Combined	**0.912**	**0.884**	**0.898**	**0.951**

suppressed; for hallucination-eliciting queries, 58.7% of sentences were suppressed and 41.3% of answers were blocked outright by the answer-level guard of (9), demonstrating fail-safe behavior precisely where it is required.

5.4 Ablation Study

Table 5 isolates each detector by using it alone as the support score (thresholds re-tuned per configuration on the development split). NLI is the strongest single signal, consistent with its unique ability to detect contradiction [54], but the full combination improves F1 by 4.2 points over NLI alone. Error analysis of the pilot runs indicates complementary failure modes: NLI alone missed out-of-domain fabrications with no meaningful premise, which cosine similarity caught; BERTScore recovered paraphrased-but-faithful sentences that NLI scored conservatively, improving precision.

5.5 Comparison With Baseline RAG

Table 6 compares against the identical retrieval-plus-generation stack with the XAI layer, detection engine, and controller removed. The framework reduced user-visible hallucination by a factor of approximately nine (18.6% $\rightarrow$ 2.1%) at the cost of $\sim$3.6 s of additional latency per query, attributable chiefly to per-sentence NLI inference.

Table 6: Proposed Framework vs. Baseline RAG (No Detection)

Metric	Baseline	Proposed
Halluc. sentences reaching user (%)	18.6	2.1
Provenance score	0.741	0.936
Answer BERTScore F1	0.868	0.917
Mean latency per query (s)	6.2	9.8
Citations per emitted sentence	0	1

6 Discussion

6.1 Strengths

The principal strength of the framework is that trustworthiness is architectural rather than advisory: unsupported content is prevented from reaching the user, every emitted sentence carries a machine-verifiable citation, and the saliency maps allow a human reviewer to audit *why* each verdict was reached—addressing the transparency gap identified in XAI surveys [25]. The multi-signal design proved materially better than any single detector because the three signals fail independently: contradiction, semantic drift, and out-of-domain fabrication are distinct phenomena, consistent with findings that combinatorial faithfulness metrics outperform individual ones [52]. Full offline operation makes the framework deployable in air-gapped environments that are categorically excluded from cloud-based verification services [23, 24], and the use of compact models keeps the entire pipeline within commodity-hardware budgets.

6.2 Limitations

Three limitations warrant acknowledgment. First, per-sentence verification adds measurable latency ($\sim$3.6 s per query in the pilot configuration); batching NLI inference and caching chunk embeddings mitigate but do not eliminate this cost. Second, detection quality is bounded by chunk quality: if inges-

tion fragments an evidentiary statement across chunk boundaries despite the 50-token overlap, a faithful sentence may be wrongly downgraded, so the false-suppression rate is partly a function of the chunking strategy. Third, the pipeline's behavior is sensitive to generator temperature; the framework was tuned at temperature 0.1, and higher temperatures increase the load on the detection engine and shift the optimal decision thresholds. Additionally, the NLI model was trained on general-domain corpora and may miscalibrate on highly specialized technical language, a limitation also noted for cross-domain NLI transfer [54, 56].

6.3 Ethical Considerations

Offline deployment is itself a privacy safeguard: sensitive documents and queries never leave the local machine, which supports compliance with data-sovereignty and confidentiality obligations. Conversely, suppression mechanisms carry a risk of over-blocking legitimate content; the transparent saliency maps and exportable audit logs are intended to keep a human meaningfully in the loop, and the system reports suppression explicitly rather than silently omitting content.

7 Conclusion and Future Work

This paper presented a fully offline, explainability-driven RAG framework in which hallucination detection is intrinsic to the generation pipeline. An XAI evidence attribution layer computes cosine attribution, chunk-level provenance, and word saliency before any answer is finalized; three independent detectors—NLI entailment, BERTScore, and cosine similarity—are fused into a weighted support score; and a tri-state controller cites, flags, or suppresses each sentence, blocking entire answers when hallucination is pervasive. The evaluation indicated strong detection performance (F1 $\approx$ 0.90, AUC-ROC $\approx$ 0.95), high retrieval quality, and an approximately ninefold reduction in user-visible hallucination relative to baseline RAG, with ablations confirming the complementarity of the three detectors.

Future work will proceed along three directions. First, the NLI cross-encoder will be fine-tuned on domain-specific entailment pairs mined from the target corpus, which is expected to improve calibration on specialized terminology [61]. Second, the detection engine will be extended to multi-hop reasoning, in which a sentence is supported only by the conjunction of multiple chunks—a gap noted across RAG hallucination studies [30, 36]. Third, the framework will be generalized to multilingual corpora by substituting multilingual sentence encoders and cross-lingual NLI models [56], enabling verified offline RAG in non-English and mixed-language document collections.

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